Palak Adnani

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PROFILE

Aspiring Robotics Engineer specialising in Engineering Product Development, with hands-on experience in mechanical design, embedded systems, electronics integration, and AI-driven solutions. Experienced in developing functional prototypes through rapid iteration, fabrication, and testing for real-world applications.

EDUCATION

Singapore University of Technology and Design (SUTD) — **Singapore**
Bachelor of Engineering- Engineering Product Development Major, Computer Science Minor — **Sep 24 - Present**

- Coursework: Circuits & Electronics, Data-Driven World, Structures & Materials, Computational and Data Driven Engineering, Design Engineering Systems, Engineering Design Innovation, Systems and Control, Electromagnetics and Applications, Modelling and Analysis

Fountainhead School — **Surat, India**
International Baccalaureate Diploma Programme (IBDP) — **Apr 22 - May 24**

- MAA HL, Physics HL, Economics HL, Chemistry SL

TECHNICAL SKILLS

Embedded Systems:
- ESP32, Raspberry Pi, Arduino, Sensors, Motor Control, Stepper Motors, RFID, PCB Design, Soldering

Programming:
- Python, C/C++, MATLAB, OpenCV, YOLO, PyTorch, HTML, CSS

Mechanical Design & Fabrication:
- Fusion 360, AutoCAD, CAD Modelling, 3D Printing, CNC Milling, Lathe, Water Jet, Zund Cutter, Welding

Robotics:
- System Integration, Prototyping, Circuit Design, Human-Centred Design

WORK EXPERIENCE

Moxin Technology (TIIDE Programme Scholarship Holder- China) — **Sep 25 - Present**
Hardware and Software Engineering Intern, Blender, Python, Stepper Motors, Electronics Integration, CAD, PCB Design — **Zhejiang University**

- Extended Hanyuan 3D deployment workflow through Blender-based image to STL generation and product customisation
- Developed a colour-mixing automation system integrating mechanical mechanisms, digital fabrication, and embedded hardware
- Extended into SUTD Baby Shark project, designing power distribution and circuit architecture for stepper motors, peristaltic pumps, and vibration mechanisms

Teacher Assistant - Technological World Course — **Jan 26 - May 26**
Electromagnetism and Circuit Analysis, TA — **Singapore**

- Assisted students for Coulomb's Law, Gauss' Law, Biot-Savart Law, Ampere's Law, Faraday's Law, and Kirchhoff's Laws in hands-on problem-solving contexts

PROJECTS

Jeeno- Assistive Feeding Claw (Robotic arm) — **May 25 - Present**
Fusion 360, CAD, Mechanical Design, Embedded Systems — **Singapore**

- Designing a human-centred 6 DOF robotic feeding assistant focused on improving independence for users with motor difficulties
- Developing mechanical structures, actuation systems, and electronics integration for robotic assistance

Underwater Robot (UWU club) — **Jan 26 - Present**
CAD, Electronics Integration, Mechanical Design — **SUTD**

- Designing internal electronics packaging for underwater robot ESC tube, optimising component placement, waterproofing considerations, and weight distribution

D.O.R.I (Drowning Onset Rescue Instrument)- 30.007 — **May 26 - Present**
CAD, Robotics, Embedded Systems — **SUTD**

- Designing robotic rescue mechanism and integrated mechanical-electrical systems for assisting drowning victims during emergency response

IRIS (Intelligent radar for Inclusive Sports) **Jan 26 - May 26**
Raspberry Pi, OpenCV, YOLO, PyTorch, Computer Vision **SUTD × ST Engineering**

- Developed a wearable assistive device enabling visually impaired football players to perceive game information through haptic feedback, integrating Raspberry Pi camera systems and YOLO-based computer vision for player/ball detection
- Integrated camera hardware with Raspberry Pi and developed image-processing pipeline for real-time detection

Global Experiential Opportunity — PV Solar Installation, **Sep 25 - Sep 25**
Site analysis, Solar PV Systems, Battery Sizing, Circuit Assembly, Data Acquisition **UGM- Indonesia**

- Worked as part of a student-faculty team to design and install a complete off-grid photovoltaic system powering a water pump in a rural village within 7 days
- Handled full electrical circuitry including battery sizing, wiring, solar charge controller integration, and on-site installation
- Received the Best Team Award

SeaSoar- Beach Cleaning Robot **Sep 24 - May 25**
ESP32, Motor Control, Electronics, CAD **Singapore**

- Integrated ESP32-based motor control system powering six motors with remote operation capability
- Developed foundational skills in soldering, wiring, and embedded systems
- Awarded Jyoti and Aditya Mathur Sustainability Award

First Tech Robotics Challenge (Centerstage) **Sep 23 - Mar 24**
Team LabFusion **Surat, Gujarat**

- Devised robot control, sensor integration, and mechanism assembly to perform tasks including object detection, color identification, and hanging mechanisms
- Team Awarded the Control Award for sensor control integration

LEADERSHIP EXPERIENCE

Namaste Indian Cultural Group, Secretary **May 25 - Jun 26**

- Managed communications, logistics, and coordination of cultural events
- Facilitated cross-cultural engagement within the university community

EXTRACURRICULAR ACTIVITIES

- Dance DerivativeZ (DDZ), Performer — Annual Production “Byte” and "Ignition"
- Badminton Club Member (Represented SUTD in SUNIG)
- HYROX Women Open — Competitive participant in functional fitness race